

AWS MODERNIZATION PLATFORM

DISASTER RECOVERY RUNBOOK

Region Failover Procedure: us-east-1 to us-west-2

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Executive Summary

This Disaster Recovery Runbook defines the procedures required to recover application services following a failure of the primary AWS Region (us-east-1).

Disaster Recovery Objectives

Recovery Time Objective (RTO): 30 Minutes

Recovery Point Objective (RPO): Less than 1 Minute

Disaster Declaration Criteria

Regional outage, critical database failure, network failure, prolonged application downtime, or a security incident requiring regional isolation.

Phase 1 – Incident Assessment

Confirm outage, review AWS Health Dashboard, verify CloudWatch alarms, assess impact, and notify stakeholders.

Phase 2 – Database Recovery

Verify DR database availability in us-west-2, promote secondary database if required, and validate connectivity.

Phase 3 – Application Recovery

Launch application instances, verify Auto Scaling Groups, and validate health checks.

Phase 4 – Kubernetes Recovery

Verify EKS cluster health, worker nodes, pods, services, and ingress configuration.

Phase 5 – Traffic Cutover

Update Route 53 DNS records and redirect traffic to the DR load balancer.

Phase 6 – Validation

Execute health checks, verify database connectivity, user authentication, and API functionality.

RTO Analysis

Estimated recovery time is approximately 30 minutes based on pre-provisioned infrastructure and automated recovery procedures.

RPO Analysis

Aurora Global Database architecture supports an estimated RPO of less than one second.

Post-Recovery Activities

Monitor stability, review metrics, conduct root cause analysis, document lessons learned, and plan failback.

Failback Procedure

Restore replication, synchronize data, validate readiness, redirect traffic back to us-east-1, and confirm operations.

Conclusion

The architecture is designed to achieve an estimated RTO of 30 minutes and an RPO of less than one second during regional failover events.